properties of light emitted and absorbed by atoms could be explained if electrons were assumed to have very particular magnetic properties. Some hundred years earlier, the Frenchman André-Marie Ampère had shown that magnetism arises from the motion of electric charge. Uhlenbeck and Goudsmit followed this lead and found that only one specific sort of electron motion could give rise to the magnetic properties suggested by the data: rotational motion—that is, spin. And so, contrary to classical expectations, Uhlenbeck and Goudsmit proclaimed that, somewhat like the earth, electrons both revolve and rotate.

Did Uhlenbeck and Goudsmit literally mean that the electron is spinning? Yes and no. What their work really showed is that there is a quantum-mechanical notion of spin that is somewhat akin to the usual image but inherently quantum mechanical in nature. It's one of those properties of the microscopic world that brushes up against classical ideas but injects an experimentally verified quantum twist. For instance, picture a spinning skater. As she pulls her arms in she spins more quickly; as she stretches out her arms she spins more slowly. And sooner or later, depending on how vigorously she threw herself into the spin, she will slow down and stop. Not so for the kind of spin revealed by Uhlenbeck and Goudsmit. According to their work and subsequent studies, every electron in the universe, always and forever, spins at one fixed and never changing rate. The spin of an electron is not a transitory state of motion as for more familiar objects that, for some reason or other, happen to be spinning. Instead, the spin of an electron is an intrinsic property, much like its mass or its electric charge. If an electron were not spinning, it would not be an electron.

Although early work focused on the electron, physicists have subsequently shown that these ideas about spin apply equally well to all of the matter particles that fill out the three families of Table 1.1. This is true down to the last detail: All of the matter particles (and their antimatter partners as well) have spin equal to that of the electron. In the language of the trade, physicists say that matter particles all have "spin-½," where the value ½ is, roughly speaking, a quantum-mechanical measure of how quickly electrons rotate.2 Moreover, physicists have shown that the nongravitational force carriers—photons, weak gauge bosons, and gluons—also possess an intrinsic